

## KALGOORLIE BOULDER, Australia

WMO: 946370

Lat: 30.7847S

Lon: 121.4533E

Elev: 366

StdP: 97.00

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 7             | 2.6        | 3.9 | -10.1                         | 1.7 | 25.6 | -7.8 | 2.0 | 24.0 | 9.8                   | 15.3 | 9.0 | 15.9 | 1.6                   | 220  | 0.486 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 1             | 14.2                   | 39.7            | 18.4 | 37.8 | 18.0 | 35.8 | 17.6 | 22.1                | 29.7 | 20.9 | 29.9 | 19.9 | 29.3 | 5.0                  | 330  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 20.1                            | 15.5 | 24.1 | 18.5 | 13.9 | 23.2 | 17.1 | 12.8 | 22.3 | 66.6          | 29.2 | 62.2 | 29.9 | 58.5 | 29.0 | 26.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| 9.3               | 8.3  | 7.6 | 0.0                        | 43.0 | 0.8                | 1.5 | -0.6                                               | 44.1 | -1.1       | 45.0 | -1.6       | 45.9 | -2.1       | 47.0 |    |    |
|                   |      |     | -0.6                       | 23.7 | 1.0                | 1.2 | -1.3                                               | 24.5 | -1.9       | 25.2 | -2.4       | 25.9 | -3.1       | 26.8 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 18.7   | 25.9  | 25.2  | 22.5  | 19.4  | 15.1  | 12.0  | 11.4  | 12.8  | 15.5  | 19.2  | 22.0  | 24.4 |
|                                                                                  | DBStd       | 6.21   | 4.12  | 4.17  | 3.99  | 3.34  | 3.11  | 2.51  | 2.49  | 3.05  | 3.51  | 4.03  | 4.14  | 4.17 |
|                                                                                  | HDD10.0     | 28     | 0     | 0     | 0     | 0     | 1     | 7     | 13    | 6     | 1     | 0     | 0     | 0    |
|                                                                                  | HDD18.3     | 873    | 1     | 1     | 7     | 27    | 109   | 189   | 216   | 173   | 99    | 38    | 12    | 2    |
|                                                                                  | CDD10.0     | 3222   | 494   | 424   | 388   | 281   | 160   | 69    | 56    | 93    | 167   | 286   | 359   | 445  |
|                                                                                  | CDD18.3     | 1024   | 236   | 192   | 137   | 57    | 9     | 1     | 0     | 3     | 14    | 65    | 121   | 189  |
|                                                                                  | CDH23.3     | 12527  | 3079  | 2346  | 1479  | 509   | 83    | 3     | 2     | 54    | 223   | 814   | 1489  | 2447 |
|                                                                                  | CDH26.7     | 6392   | 1755  | 1284  | 701   | 170   | 13    | 0     | 0     | 8     | 64    | 342   | 724   | 1331 |
| Wind                                                                             | WSAvg       | 4.0    | 4.9   | 4.7   | 4.2   | 3.4   | 3.0   | 3.1   | 3.3   | 3.5   | 3.9   | 4.4   | 4.6   | 4.8  |
| Precipitation                                                                    | PrecAvg     | 275    | 27    | 32    | 29    | 22    | 27    | 28    | 26    | 22    | 14    | 16    | 20    | 18   |
|                                                                                  | PrecMax     | 530    | 170   | 197   | 199   | 99    | 110   | 186   | 83    | 74    | 59    | 84    | 115   | 89   |
|                                                                                  | PrecMin     | 124    | 0     | 0     | 0     | 0     | 0     | 1     | 0     | 2     | 0     | 0     | 0     | 0    |
|                                                                                  | PrecStd     | 99     | 34    | 38    | 40    | 23    | 26    | 30    | 19    | 17    | 12    | 19    | 20    | 18   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 42.7  | 41.8  | 38.7  | 33.9  | 28.4  | 23.2  | 23.1  | 27.9  | 31.4  | 35.6  | 38.9  | 40.6 |
|                                                                                  |             | MCWB   | 18.6  | 19.0  | 17.4  | 16.2  | 14.2  | 12.2  | 11.3  | 12.5  | 14.2  | 15.3  | 16.9  | 17.7 |
|                                                                                  | 2%          | DB     | 39.9  | 38.9  | 35.7  | 31.0  | 25.7  | 21.0  | 20.9  | 24.7  | 28.4  | 33.1  | 36.0  | 38.4 |
|                                                                                  |             | MCWB   | 18.6  | 19.0  | 17.4  | 15.6  | 13.9  | 11.7  | 10.9  | 11.6  | 13.2  | 14.9  | 16.4  | 17.7 |
|                                                                                  | 5%          | DB     | 37.5  | 36.4  | 33.1  | 28.3  | 23.5  | 19.1  | 19.0  | 22.0  | 25.9  | 30.8  | 33.3  | 36.1 |
|                                                                                  |             | MCWB   | 18.5  | 18.6  | 17.4  | 15.5  | 13.4  | 11.1  | 10.5  | 11.2  | 12.7  | 14.5  | 15.8  | 17.4 |
|                                                                                  | 10%         | DB     | 35.0  | 34.0  | 30.7  | 26.1  | 21.4  | 17.6  | 17.2  | 19.6  | 23.4  | 28.0  | 31.0  | 33.7 |
|                                                                                  |             | MCWB   | 18.0  | 18.2  | 17.0  | 15.2  | 12.9  | 10.6  | 10.2  | 10.6  | 12.1  | 13.9  | 15.3  | 16.9 |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 23.3  | 24.0  | 21.3  | 19.6  | 17.3  | 15.0  | 14.4  | 14.8  | 16.2  | 18.3  | 19.5  | 21.6 |
|                                                                                  |             | MCDB   | 30.0  | 30.1  | 27.8  | 26.1  | 22.4  | 19.2  | 17.8  | 19.9  | 25.5  | 28.2  | 28.5  | 31.5 |
|                                                                                  | 2%          | WB     | 21.8  | 22.6  | 20.0  | 18.2  | 16.1  | 13.7  | 12.9  | 13.5  | 14.6  | 16.7  | 18.3  | 20.2 |
|                                                                                  |             | MCDB   | 30.9  | 29.3  | 27.4  | 25.2  | 20.6  | 17.5  | 17.1  | 20.0  | 24.3  | 26.3  | 29.3  | 30.3 |
|                                                                                  | 5%          | WB     | 20.8  | 21.2  | 19.1  | 17.3  | 15.0  | 12.6  | 11.9  | 12.4  | 13.7  | 15.7  | 17.4  | 19.0 |
|                                                                                  |             | MCDB   | 30.7  | 29.2  | 27.1  | 23.6  | 20.3  | 16.6  | 16.4  | 19.1  | 22.7  | 26.5  | 28.6  | 30.6 |
|                                                                                  | 10%         | WB     | 19.6  | 20.0  | 18.2  | 16.4  | 14.0  | 11.6  | 11.0  | 11.5  | 12.9  | 14.7  | 16.5  | 17.9 |
|                                                                                  |             | MCDB   | 30.5  | 29.2  | 26.5  | 23.1  | 19.6  | 15.8  | 15.8  | 18.1  | 21.2  | 25.7  | 27.8  | 30.2 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 14.2  | 13.1  | 12.1  | 11.9  | 11.3  | 10.9  | 11.1  | 12.4  | 13.4  | 13.9  | 13.9  | 14.3 |
|                                                                                  |             | MCDBR  | 17.2  | 16.7  | 15.6  | 14.7  | 13.7  | 12.7  | 13.8  | 15.6  | 17.0  | 17.1  | 17.1  | 17.0 |
|                                                                                  |             | MCWBR  | 5.1   | 4.9   | 5.1   | 5.2   | 5.8   | 6.2   | 6.6   | 6.8   | 6.6   | 6.0   | 5.4   | 5.3  |
|                                                                                  | 5% WB       | MCDBR  | 14.5  | 13.3  | 12.1  | 11.6  | 10.4  | 10.2  | 10.7  | 12.6  | 14.1  | 14.7  | 15.3  | 15.1 |
|                                                                                  |             | MCWBR  | 5.1   | 5.1   | 4.9   | 5.0   | 5.2   | 5.8   | 6.0   | 6.5   | 6.1   | 5.7   | 5.2   | 5.5  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.334  | 0.337 | 0.317 | 0.304 | 0.286 | 0.276 | 0.274 | 0.282 | 0.300 | 0.315 | 0.321 | 0.321 |      |
|                                                                                  | taud        | 2.605  | 2.615 | 2.650 | 2.659 | 2.687 | 2.688 | 2.695 | 2.642 | 2.579 | 2.554 | 2.567 | 2.604 |      |
|                                                                                  | Ebn at Noon | 1006   | 985   | 974   | 937   | 911   | 897   | 916   | 948   | 975   | 996   | 1012  | 1022  |      |
|                                                                                  | Edh at Noon | 103    | 99    | 90    | 80    | 70    | 66    | 68    | 80    | 95    | 104   | 107   | 104   |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 7.78   | 6.83  | 5.88  | 4.61  | 3.55  | 2.99  | 3.27  | 4.28  | 5.66  | 6.82  | 7.61  | 8.16  |      |
|                                                                                  | RadStd      | 0.53   | 0.61  | 0.41  | 0.26  | 0.24  | 0.18  | 0.18  | 0.20  | 0.25  | 0.40  | 0.39  | 0.46  |      |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (d)  | (e)                    | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)     |         |         |
| (46) | Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (2 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page